Gateways, Load Balancers, and Observability with k6

COMPSCI 426

- Reverse proxies, gateways, load balancers
- Caddy and k6
- Comparing system behavior under load

Today

1. Concepts
2. Tools
3. Scenario architecture preview
4. Reading k6 output
5. Debrief

Learning Goals

By the end of class, you should be able to:

- explain gateways and load balancers
- configure Caddy as a reverse proxy
- run and interpret a k6 test
- compare deployment setups under load

The Core Question

> You scaled your service to 3 replicas. Great. Who decides where the next request goes?

The Problem Setup

You scaled your service to 3 replicas.

Now what?

- Who decides which replica gets the next request?
- How does outside traffic find the right place?
- How would you know whether the setup is working?

Theme: architecture choices should create measurable effects.

Reverse Proxy vs Forward Proxy

Reverse proxy

- sits in front of servers
- forwards traffic to backends
- hides service topology

Forward proxy

- sits in front of clients
- represents clients to the outside world
- often used for filtering or privacy

Shortcut: reverse proxy protects servers; forward proxy represents clients.

What a Gateway Does

A gateway is the **front door** of your system.

- routes requests
- terminates TLS
- can enforce auth
- can enforce rate limits
- centralizes policy

Every outside request usually touches it first.

Where Load Balancing Fits

A load balancer is one key concern inside or alongside a gateway.

- spreads traffic across replicas
- prevents one instance from taking all the heat
- improves throughput
- improves latency consistency

Without routing, extra replicas may exist but still do no work.

Common Strategies

- **Round robin**: take turns across replicas
- **Least connections**: prefer the least busy backend
- **IP hash**: send the same client toward the same backend

The question to keep asking:

How would you know whether the strategy is actually working?

Why This Matters Here

This applies directly to what you have built:

- scaled services need smart traffic flow
- pub-sub systems benefit from gateway rate limits
- resilient workers still depend on a stable entry point
- projects need evidence, not guesses, about load behavior

Meet Caddy

```
:80 {
    reverse_proxy service:3000 {
        lb_policy round_robin
    }
}
```

- readable config
- small surface area
- easy to demo in Docker

Same ideas transfer to nginx and cloud load balancers.

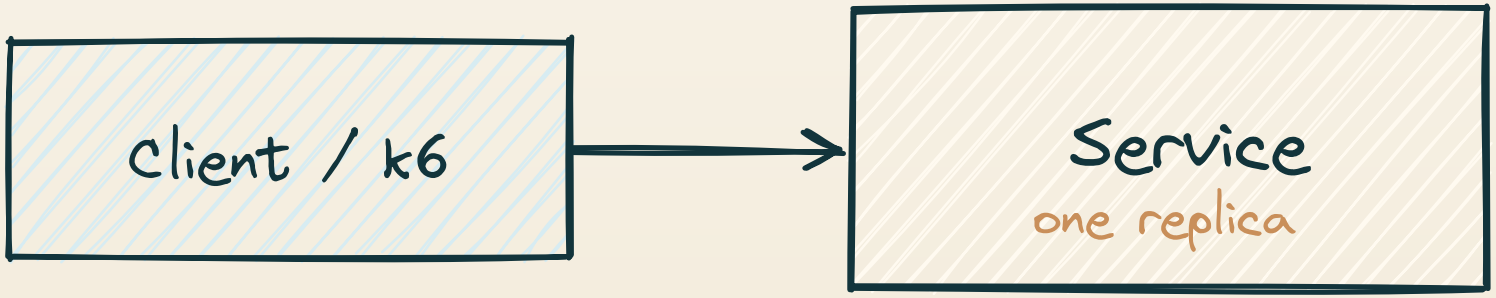
Meet k6

k6 is a developer-oriented load testing tool.

- scripts are JavaScript
- VUs model concurrent users
- duration controls how long the test runs
- reports throughput, latency percentiles, and failures

Important: k6 tells you what changed, not automatically why.

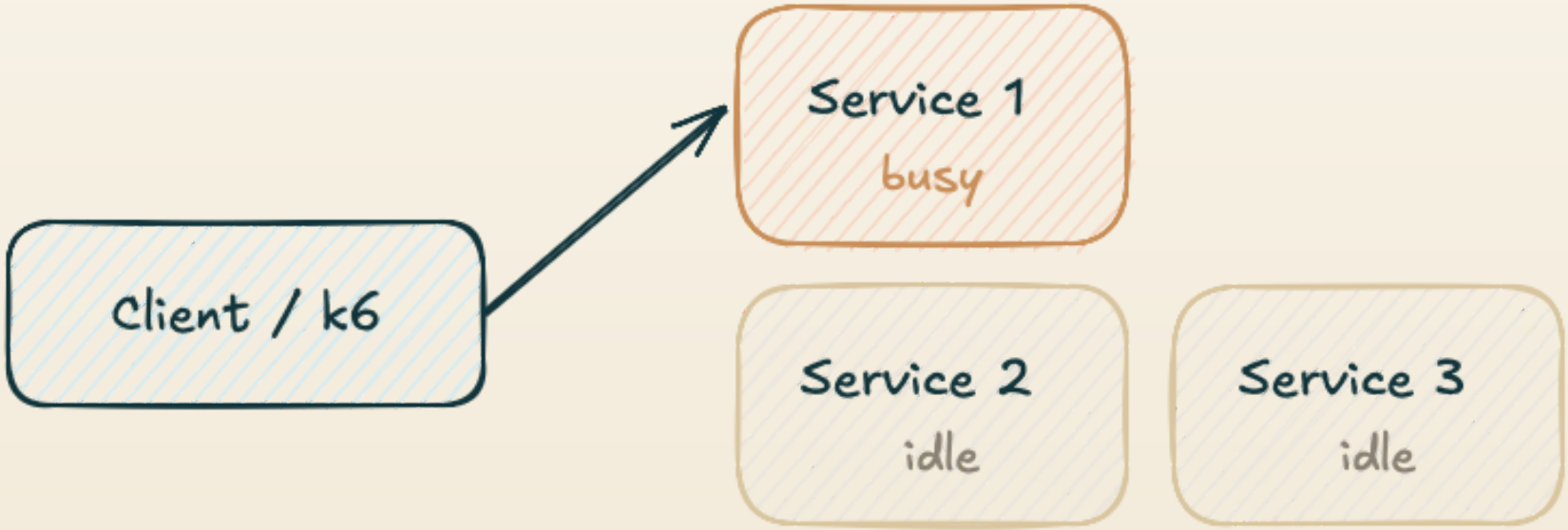
Scenario 1 Architecture



- one replica
- no gateway
- no load balancer
- direct hit on one service

What you learn: baseline behavior under concurrent traffic.

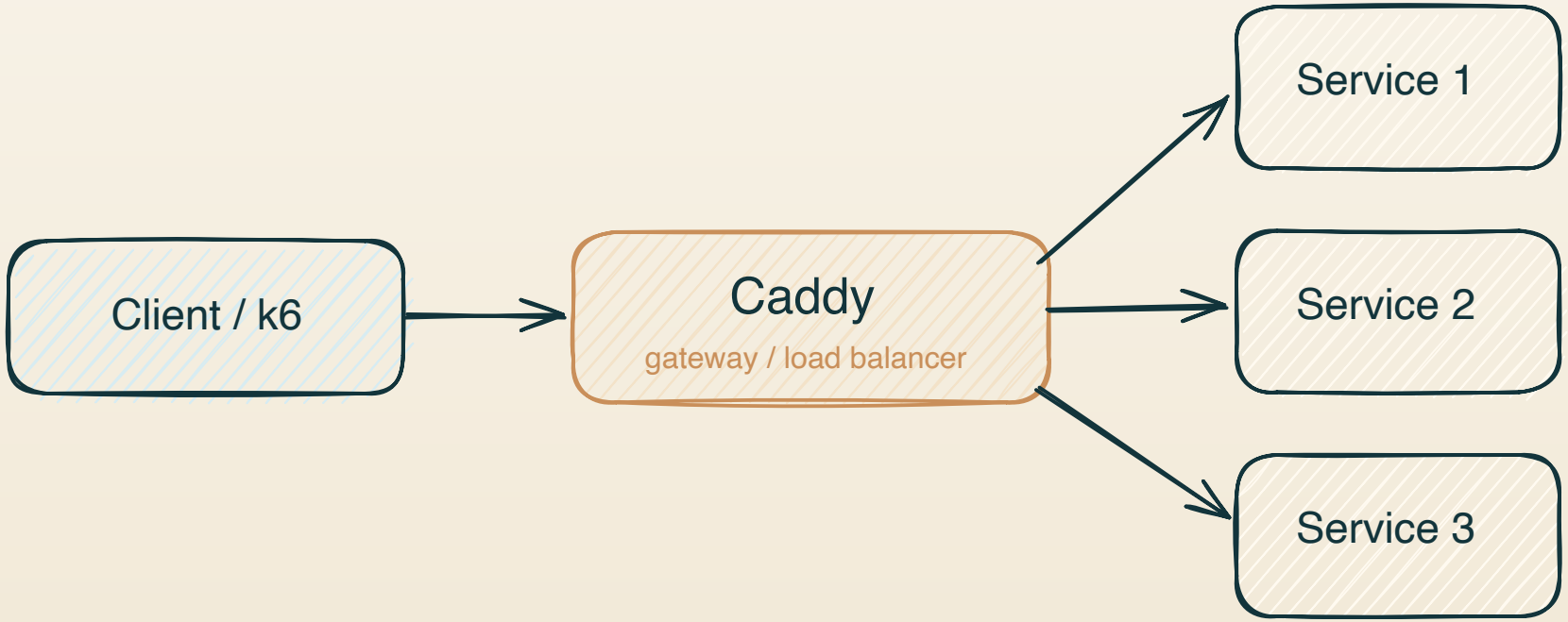
Scenario 2 Architecture



- three replicas exist
- traffic still hits only one replica
- architecture changed, traffic pattern did not

What you learn: scaling alone is not enough.

Scenario 3 Architecture



- one front door
- requests distributed across replicas

What you learn: routing changes measured system behavior.

One Workload, Three Traffic Paths

- Scenario 1: one service, direct traffic
- Scenario 2: three services, but one still gets the load
- Scenario 3: one front door, traffic distributed across replicas

What stays constant:

- same app shape
- same style of workload

What changes:

- the traffic path through the system

k6 Output: First Pass

```
scenarios: (100.00%) 1 scenario, 20 max VUs
default: 20 looping VUs for 30s

http_req_duration: avg=142ms p(95)=244ms
http_req_failed: 0.00%
http_reqs: 423  14.08/s
iteration_duration: avg=1.14s
vus: 20
```

Read it in this order:

1. concurrency
2. latency
3. failures
4. throughput

k6 Output: What Matters

- **p95 vs average**
  - average can hide bad tail behavior
- `http_req_failed`
  - tells you that requests failed, not why
- **iterations vs requests**
  - one loop can make multiple requests
- **VUs vs requests/sec**
  - concurrency and throughput are related, not identical

How to Talk About Results

Push past “it was faster.”

Ask:

- which metric changed?
- by how much?
- was the change consistent?
- what architecture change explains it?
- what should you inspect next?

Use k6 to guide investigation, not to end it.

Debrief

This connects back to the course arc:

- you can now verify resilience under load
- you have language for front-door traffic control
- you can reason about whether a project handles real traffic
- you can compare architectures using evidence

Closing Question

If your gateway goes down, what happens to your entire system?

That question leads directly to:

- single points of failure
- high availability gateways
- project-level reliability thinking